

Image Processing Assessment

Bone Segmentation

My first step for the bone segmentation task was to load the CT scan from the nibabel library. the image was thresholded to separate the bony structures based on Hounsfield Units (HU). Since I reviewed the range of values in the image, I set 320 HU as the threshold for bone scenarios in the data. As a result, skeleton regions were labeled as 1 and everything else as 0.

Key Steps

1. Loaded the CT Scan in .nii.gz format.
2. Threshold was set to 320 HU.
3. Mask of the segmented bone regions created that uses 0s and 1s.
4. Saved the segmented bone mask under a new .nii.gz format.

Contour Expansion

To carry out contour expansion, I added a uniform 2 mm to each side of the segmented bone mask. This was done by:

1. Using the image header information to measure where each of the 3D voxel points should be placed.
2. Forming a round structuring element (kernel) so that every voxel within 2 mm from the center voxel is included.
3. Using this structuring element to expand pixels by one unit with binary dilation.

The outcome is a new mask that increases the bone segmentation by 2 mm. I kept this expansion in a separate .nii.gz file. I made the expansion radius a parameter so that it's easy to change if necessary.

Randomized Contour Adjustment

For this task, I implemented a randomized contour adjustment between the original segmentation and the expanded mask. The steps included the following:

1. First, I found the area referred to as the expansion region (those pixels included by the 2 mm expansion that were not included in the original testing).
2. After that, I randomly chose some voxels from this region to put into the final mask. Because of this randomness, we can generate several types of contours for spreading data.

This approach managed to ensure that:

1. None of the new data points was outside the 2 mm expanded zone.
2. Each mask after adjustment still contained the original edges of the bone.

The file containing the randomized adjusted mask was saved as .nii.gz.

Landmark Detection on Tibia

1. The landmark detection task tries to extract the medial and lateral lowest points on the tibial surface. The steps included the following:
2. First of all, the tibial bone mask was extracted.
3. I converted the voxel positions of the tibial mask to their world coordinates with the help of the image affine.
4. Points were taken from the tibia that lay at coordinates among the tibial plateau (upper 20%) in the z-axis.
5. Taking these observations, I selected the 10% with the lowest z values to be my candidates for landmarks.
6. Then the points were split into medial and lateral groups based on the median of their x-coordinates.

The identified medial and lateral lowest points on the tibial plateau were recorded for each of the 5 masks:

1. Original mask

The extracted coordinates are:

- Medial lowest point: [-114.73, -10.43, -590.50]
- Lateral lowest point: [-111.25, -9.56, -590.50]

2. 2 mm expanded mask.

The extracted coordinates are:

- Medial lowest point: [-129.50, -112.99, -602.50]
- Lateral lowest point: [-96.47, -166.01, -610.50]

3. 4 mm expanded mask.

The extracted coordinates are:

- Medial lowest point: [-128.63, -114.73, -602.50]
- Lateral lowest point: [-95.61, -167.74, -610.50]

4. Randomized mask 1

The extracted coordinates are:

- Medial lowest point: [-110.38, -117.33, -596.50]
- Lateral lowest point: [-94.74, -166.88, -602.50]

5. Randomized mask 2

The extracted coordinates are:

- Medial lowest point: [-109.51, -118.20, -596.50]
- Lateral lowest point: [-95.61, -166.88, -594.50]